

Boundary slope is irrelevant for the channel Poincaré constant

Automated mathematical discovery run `fa_banach_001`

13 August 2026

Status. Candidate full solution to source Remark B.2, likely valid, subject to human review. The mean-zero Poincaré constant of the periodic channel has an explicit bound independent of the boundary slope. An additional oscillatory example proves square-root lower growth in the geometry parameter for the distinct Stokes inf-sup problem.

1 Source question

The source is J. Wilkening, *Inf-sup estimates for the Stokes problem in a periodic channel*, arXiv:0706.4082 [1]. Let $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$ and

$$\Omega_h = \{(x, y) : x \in \mathbb{T}_L, 0 < y < h(x)\}, \quad 0 < h_0 \leq h(x) \leq h_1.$$

The source proves, for mean-zero $p \in H^1(\Omega_h)$, $L^{-1}\|p\|_2 \leq C\|\nabla p\|_2$ with a constant containing the factor $1 + M$, where $M = \|h'\|_\infty$. Remark B.2 on source p. 18 asks whether this linear dependence on M is optimal. The exact passage is reproduced below.

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Next, we choose $H = \sqrt{h_0 h_1}$ and minimize $\max(\gamma_1, 1 + \gamma_2 M^2)$ over all choices of γ_j such that $\gamma_1^{-1} + \gamma_2^{-1} = 1$. The result is

$$\gamma_1 = 1 + \gamma_2 M^2 = \frac{1}{4}(\sqrt{M^2 + 4} + M)^2 \leq (1 + M)^2, \quad (\text{B.11})$$

which yields $C^2 = \frac{1}{4\pi^2} \max\left(1, 4\frac{h_0 h_1}{L^2}\right) \frac{h_1}{h_0} (1 + M)^2$ as claimed. $\square$

REMARK B.2. Example 4.5 shows that the scaling $C \sim h_0^{-1/2}$ is optimal: for that function p , we have

$$L^{-2}\|p\|_{0,\Omega}^2 \geq L^{-2}h_1 \frac{L}{4} = \frac{h_1}{256h_0} \left(h_0 \frac{L}{4} \left(\frac{16}{L}\right)^2\right) \geq \frac{h_1}{256h_0} \|p_x\|_{0,\Omega}^2, \quad (\text{B.12})$$

which shows that C in (B.3) is at least $\frac{1}{16}\sqrt{\frac{h_1}{h_0}}$. We do not know if the linear dependence of C on M is optimal — it seems to be an unavoidable artifact of changing variables to a rectangular geometry.

The answer is no: no dependence on M is needed. The packet also treats the separate Remark 4.3, which asks whether quadratic dependence on a $C^{1,1}$ geometry parameter is optimal for the Stokes inf-sup inverse.

2 Proof intuition

Every channel in the class contains the same flat base rectangle $R_0 = \mathbb{T}_L \times (0, h_0)$. Anchor p to its average on R_0 , where the exact rectangular Poincaré estimate is available. An averaged

one-dimensional identity controls the trace at height h_0 . Values above that height are then reached by integrating only in the vertical direction. None of these operations differentiates h , so the boundary slope never appears.

The Stokes inf-sup constant behaves differently. A rapidly oscillating upper boundary creates many upper teeth. Put alternating signs on those teeth. The divergence pairing collapses to an alternating trace functional on the common interface $y = h_0$; its $H^{-1/2}$ norm is of order $N^{-1/2}$. This forces the inf-sup inverse to grow like $\sqrt{N}$, and hence like the square root of the source's geometry parameter.

3 A slope-free Poincaré theorem

Set

$$D = h_1 - h_0, \quad \lambda_0 = \max \left\{ \frac{L}{2\pi}, \frac{h_0}{\pi} \right\},$$

and define

$$K_0^2 = \max \left\{ \left(1 + \frac{4D}{h_0} \right) \lambda_0^2 + \frac{4Dh_0}{3}, D^2 \right\}.$$

Theorem 1 (Slope-free mean-zero estimate). *Let $h \in C^{0,1}(\mathbb{T}_L)$ satisfy $h_0 \leq h \leq h_1$. If $p \in H^1(\Omega_h)$ and $\int_{\Omega_h} p = 0$, then*

$$\|p\|_{L^2(\Omega_h)} \leq K_0 \|\nabla p\|_{L^2(\Omega_h)}.$$

In particular, K_0 depends only on L, h_0, h_1 , not on $\|h'\|_\infty$.

Proof. Let $R_0 = \mathbb{T}_L \times (0, h_0)$, let c be the average of p on R_0 , and put $g = p - c$. Fourier expansion on the periodic rectangle, with Neumann conditions implicit in the variational inequality, gives

$$\|g\|_{L^2(R_0)} \leq \lambda_0 \|\nabla p\|_{L^2(R_0)}. \quad (1)$$

For almost every x , one-dimensional absolute continuity and averaging in the lower fiber give the exact identity

$$g(x, h_0) = \frac{1}{h_0} \int_0^{h_0} g(x, s) ds + \frac{1}{h_0} \int_0^{h_0} t p_y(x, t) dt.$$

Consequently,

$$\int_{\mathbb{T}_L} |g(x, h_0)|^2 dx \leq \frac{2}{h_0} \|g\|_{L^2(R_0)}^2 + \frac{2h_0}{3} \|p_y\|_{L^2(R_0)}^2. \quad (2)$$

Let $U = \Omega_h \setminus R_0$. For $h_0 < y < h(x)$, $g(x, y) = g(x, h_0) + \int_{h_0}^y p_y(x, t) dt$. Integrating in y , interchanging the order in the second term, and using $0 \leq h(x) - h_0 \leq D$, we obtain

$$\|g\|_{L^2(U)}^2 \leq 2D \int_{\mathbb{T}_L} |g(x, h_0)|^2 dx + D^2 \|p_y\|_{L^2(U)}^2. \quad (3)$$

Equations (1)–(3) yield

$$\|p - c\|_{L^2(\Omega_h)}^2 \leq K_0^2 \|\nabla p\|_{L^2(\Omega_h)}^2.$$

Finally, the whole-channel mean of p is zero, so

$$\|p - c\|_{L^2(\Omega_h)}^2 = \|p\|_{L^2(\Omega_h)}^2 + |\Omega_h| c^2 \geq \|p\|_{L^2(\Omega_h)}^2.$$

This proves the theorem. The fiber calculation for general H^1 functions follows by the standard density and slicing properties on a Lipschitz subgraph. $\square$

Corollary 2 (Answer to Remark B.2). *In the normalization of source Theorem B.1 one may take*

$$C = K_0/L,$$

which is independent of $M = \|h'\|_\infty$. Hence the linear M -dependence in the source estimate is entirely an artifact of its boundary-flattening proof.

4 Oscillatory lower bound for the Stokes inf–sup inverse

For a periodic channel Ω , write

$$\beta(\Omega) = \inf_{\substack{q \in L^2(\Omega), \int q = 0 \\ q \neq 0}} \sup_{\substack{\mathbf{u} \in H_0^1(\Omega)^2 \\ \mathbf{u} \neq 0}} \frac{|\int_{\Omega} q \operatorname{div} \mathbf{u}|}{\|q\|_2 \|\nabla \mathbf{u}\|_2}.$$

Here the zero trace is imposed on the top and bottom walls, with periodicity in x , as in the source.

Theorem 3 (Alternating-tooth obstruction). *Fix $h_0, D > 0$. For every positive even integer N , let*

$$h_N(x) = h_0 + D(1 - \cos Nx), \quad x \in \mathbb{T}_{2\pi},$$

and let $\Omega_N = \Omega_{h_N}$. Then

$$\beta(\Omega_N) \leq \left(\frac{14\zeta(3)}{\pi^2 DN} \right)^{1/2}, \quad \beta(\Omega_N)^{-1} \geq \pi \left(\frac{DN}{14\zeta(3)} \right)^{1/2}.$$

If M_N is the geometry parameter of source Theorem 4.1, then $M_N = \kappa N$ for a fixed $\kappa = \kappa(h_0, D) > 0$. Thus $\beta(\Omega_N)^{-1} \geq c(h_0, D)\sqrt{M_N}$.

Proof. The region above $y = h_0$ has N teeth

$$A_j = \left\{ \frac{2\pi j}{N} < x < \frac{2\pi(j+1)}{N}, \quad h_0 < y < h_N(x) \right\}, \quad 0 \leq j < N.$$

Let $q_N = (-1)^j$ on A_j and $q_N = 0$ on the base rectangle. All teeth have the same area and N is even, hence $\int q_N = 0$, while

$$\|q_N\|_2^2 = \sum_j |A_j| = 2\pi D. \quad (4)$$

For $\mathbf{u} \in H_0^1(\Omega_N)^2$, integration on each tooth uses the zero trace on the curved top. The horizontal flux integrates to zero at the endpoints, where the tooth height vanishes, and therefore

$$-\int_{\Omega_N} q_N \operatorname{div} \mathbf{u} = \int_0^{2\pi} w_N(x) u_2(x, h_0) dx, \quad (5)$$

where w_N alternates between $+1$ and -1 on consecutive cells. Equivalently, $w_N(x) = \operatorname{sgn}(\sin(Nx/2))$.

With the Fourier convention $\widehat{w}(k) = (2\pi)^{-1} \int_0^{2\pi} w(x) e^{-ikx} dx$, the square-wave expansion gives nonzero modes only at $k = (2m+1)N/2$, and

$$\sum_{k \neq 0} \frac{|\widehat{w}_N(k)|^2}{|k|} = \frac{16}{\pi^2 N} \sum_{m=0}^{\infty} \frac{1}{(2m+1)^3} = \frac{14\zeta(3)}{\pi^2 N}. \quad (6)$$

Put $v(x) = u_2(x, h_0)$. For every nonzero horizontal mode,

$$\int_0^{h_0} (|f'|^2 + k^2 |f|^2) dy \geq |k| |f(h_0)|^2 \quad (f(0) = 0),$$

as follows by expanding $\int_0^{h_0} |f' - |k|f|^2$. Parseval therefore gives

$$2\pi \sum_{k \neq 0} |k| |\widehat{v}(k)|^2 \leq \|\nabla u_2\|_{L^2(\mathbb{T}_{2\pi} \times (0, h_0))}^2 \leq \|\nabla \mathbf{u}\|_{L^2(\Omega_N)}^2. \quad (7)$$

Cauchy–Schwarz in (5), followed by (6)–(7), yields

$$\left| \int_{\Omega_N} q_N \operatorname{div} \mathbf{u} \right| \leq \left(\frac{28\zeta(3)}{\pi N} \right)^{1/2} \|\nabla \mathbf{u}\|_2.$$

Divide by (4) and use q_N as a competitor in the definition of β .

Finally, $\|h'_N\|_\infty = DN$, and $\frac{1}{2}\|h_N h''_N\|_\infty$ is a fixed positive multiple of N^2 . Hence the source parameter

$$M_N^2 = \max \left\{ \|h'_N\|_\infty^2, \frac{1}{2} \|h_N h''_N\|_\infty \right\}$$

equals $\kappa^2 N^2$ for fixed $\kappa > 0$. □

5 Verification, limitations, and novelty

Theorem 1 fully answers the extracted Remark B.2. It does not claim the displayed constant is sharp in L, h_0, h_1 . Theorem 3 is a substantial partial result for the distinct Remark 4.3: the source supplies an $O(M^2)$ upper bound, while the new family forces $\Omega(\sqrt{M})$ growth. The exponent between $1/2$ and 2 remains open.

Eight focused checks and upgrade attempts covered the trace coefficient, the two different means, Sobolev slicing, exact square-wave normalization, zero-mode removal, tooth flux identity, a direct vertical divergence construction, Piola flattening, and quantitative Bogovskiĭ/John-domain estimates. The latter routes reintroduced uncontrolled horizontal derivatives or curvature and did not justify a universal subquadratic upper bound.

A bounded novelty search used the exact title, arXiv id, author, both quoted optimality phrases, and combinations of “periodic channel”, “Poincaré constant”, “LBB constant”, “slope”, and “oscillating boundary”. It also checked general uniform-Poincaré work [3] and quantitative Bogovskiĭ estimates [2, 4]. No explicit answer to Remark B.2, no base-rectangle proof, and no alternating-tooth lower bound for Remark 4.3 were found. Novelty confidence is provisional pending expert review.

References

- [1] J. Wilkening, *Inf-sup estimates for the Stokes problem in a periodic channel*, arXiv:0706.4082 (2007), especially Remarks 4.3 and B.2.
- [2] R. G. Durán, *An elementary proof of the continuity from $L_0^2(\Omega)$ to $H_0^1(\Omega)^n$ of Bogovskiĭ’s right inverse of the divergence*, arXiv:1103.3718 (2011).
- [3] D. Ruiz, *A note on the uniformity of the constant in the Poincaré inequality*, arXiv:1208.6045 (2012).
- [4] J. Guzmán and A. J. Salgado, *Estimation of the continuity constants for Bogovskiĭ and regularized Poincaré integral operators*, arXiv:2010.04105 (2020).